The D-Family Transport Strike: the Weil-world package at every squarefree d (Round 40)

GST Hodge campaign – research artifact

2026-11-06 (session clock)

Scope and the standing order

Operator order (the standing finish order + the continuation order + this session's order: full scope, both skills live, no stopping): SKILL V6.2 pre-read in full (752 lines, §0–§46); the ledger rehearsed through 113; the PTS repo (`prove-that-shit`) cloned live and its *prove-open-problem* and *adversarial-proof-audit* skills engaged side-by-side (the adversarial-auditor role discharged by the AUDIT-40 agent). Target: the **transport families beyond the censuses** – Round 39's closing line named them as the first item of the remaining falsifiable line. This round executes the family on the Weil face: the Round 38/39 package transported from the nine Heegner seeds to *every* squarefree d, with the exact exclusion collar computed, and the load-bearing arithmetic laws certified as polynomial identities in the symbol d.

1 The family and the phase classification

Throughout, $K = \mathbb{Q}(\sqrt{-d})$ with $d \geq 1$ squarefree, and $\alpha = a + b\sqrt{-d} \in \mathbb{Z}[\sqrt{-d}]$; the letter model (8 letters e_j, f_j, $j = 1..4$; $[\alpha]^*$ the pullback/row convention; $\theta = \sum_j e_j \wedge f_j$; the 70 transversal-and-mixed words) is Round 38's, unchanged. Round 38's census committed the nine Heegner-class pairs (d, α); the round's conservative genericity criterion $(\alpha/\bar{\alpha})^k \neq 1$ for $k \leq 4$ excluded all root-of-unity phases.

Theorem 1.1 (S29, the phase classification and the exact collar)**.** *Write $\zeta_\alpha = \alpha/\bar{\alpha}$ (a norm-one element of K). The roots of unity of an imaginary quadratic field have order at most 6, and only $\mathbb{Q}(i)$ (order 4) and $\mathbb{Q}(\sqrt{-3})$ (orders 3 and 6) carry any beyond ± 1. Consequently:*

(i) ζ_α has finite order only for $\alpha/\bar{\alpha} \in \mu(K)$, and the order lies in $\{1, 2, 3, 4, 6\}$;

(ii) order 1 $\iff$ $b = 0$; order 2 $\iff$ $a = 0$; order 3 $\iff$ $d = 3$ and $a = \pm b$; order 4 $\iff$ $d = 1$ and $a = \pm b$; order 6 $\iff$ $d = 3$ and $a = \pm 3b$;

*(iii) the **exact exclusion collar** of the family is*

$$\{b = 0\} \cup \{a = 0\} \cup \{(d, a, b) : d = 3,\ a = \pm b\} \quad \text{(the phase orders 1, 2, 3),}$$

*and the **order-4 and order-6 phase worlds are family members**: at these worlds the spectral law (dims 2/36/32, pairwise coprimality) still holds – only the Weil factor's* degree *degenerates at order 4 (*minpoly(α^4) *linear, $\alpha^4 \in \mathbb{Q}$), which does not affect any load-bearing receipt.*

Proof. (i) is the classical torsion theorem for imaginary quadratic unit groups. For (ii): $\zeta_\alpha = 1$ forces $\alpha = \bar{\alpha}$, i.e. $b = 0$; $\zeta_\alpha = -1$ forces $a = 0$. Order 3 or 6 requires $\zeta_\alpha \in \mathbb{Q}(\sqrt{-3})^\times$, hence $d = 3$; there $\zeta = \frac{(a^2-3b^2)+2ab\sqrt{-3}}{a^2+3b^2}$ equals $e^{\pm 2\pi i/3}$ exactly when $a = \pm b$ (order 3) and equals $e^{\pm \pi i/3}$ exactly when $a = \pm 3b$ (order 6). Order 4 requires $d = 1$ and $\zeta_\alpha = \pm i$, i.e. $a = \pm b$. For (iii): the spectral factors are $\mathrm{minpoly}(\alpha^4)$, $\mathrm{minpoly}(\alpha^3\bar{\alpha})$, $t - N(\alpha)^2$; shared roots occur exactly when $\alpha^4 \in \{\alpha^3\bar{\alpha}\text{-orbit}, N(\alpha)^2\}$, i.e. when $\zeta_\alpha^3 = 1$, $\zeta_\alpha^2 = 1$, or $\zeta_\alpha = 1$ (the orders 3, 2, 1). At order 4: $\alpha^4 = \bar{\alpha}^4 \in \mathbb{Q}$, the Weil factor is linear with root distinct from $N(\alpha)^2$ (which has order-2 phase) and from the mixed factor's roots (which carry order 3); the three factors remain pairwise coprime, the α^4-primary part retains dimension $2 = \binom{4}{4} + \binom{4}{0}$, and every receipt of Rounds 38/39 applies verbatim. At order 6 no sharing condition fires at all. The certificate verifies the classification computationally at a 160-instance (d, a, b) grid, both directions, with the collar worlds detected by the certificate's own coprime machinery. □

Corollary 1.2 (The family census)**.** *The family*

$$\mathcal{F} \;=\; \{(d, a, b) : d \text{ squarefree},\ ab \neq 0,\ (d, a, b) \notin \text{collar}\}$$

is infinite. The committed census of this round: every squarefree $d \leq 45$ plus the Heegner seeds $67, 163$, at the $\alpha = 1 + \sqrt{-d}$ and $\alpha = 2 + \sqrt{-d}$ faces, plus the order-4 member $(1, 1 + i)$ and the order-6 member $(3, 3 + \sqrt{-3})$: 62 worlds, covering all nine Round-38/39 seeds and 22 non-Heegner d's.

2 The d-uniform world-interface law

Theorem 2.1 (S30, the spectral law and the integral Weil lattice at the family)**.** *At every family world (d, α):*

(i) the $\mathbb{Q}$-spectral structure of $[\alpha]^*$ on $\bigwedge^4_{\mathbb{Q}} H^1$ is $\mathrm{minpoly}(\alpha^4)^1 \oplus \mathrm{minpoly}(\alpha^3\bar{\alpha})^{16} \oplus (t - N(\alpha)^2)^{36}$ with $2 + 32 + 36 = 70$ (trace-verified against the actual 70×70 exterior matrix); the Weil plane W = the α^4-primary part has dimension exactly 2;

(ii) the integral Weil classes

$$\hat{w}_R = \sum_{t \text{ even}} (1, -d, d^2)\, w_T, \qquad \hat{w}_I = \sum_{t \text{ odd}} (1, -d)\, w_T$$

(the 16 transversal words) are integral, primitive, lie in W, and $W \cap \mathbb{Z}^{70} = \langle \hat{w}_R, \hat{w}_I \rangle_{\mathbb{Z}}$ (the $t = 0/t = 1$ coordinate-forcing);

(iii) the periods $\mathrm{INT}\, \hat{w}_R^2 = 8d^2$, $\mathrm{INT}\, \hat{w}_I^2 = 8d$, $\mathrm{INT}\, \hat{w}_R \hat{w}_I = 0$, $\mathrm{INT}\, \hat{w}_R \theta^2 = 0$, $\mathrm{INT}\, \theta^4 = 24$;

(iv) the CRT projector (the unique degree-≤ 4 $\mathbb{Q}$-polynomial $P \equiv 1$ mod the Weil factor, 0 mod the others) is an explicit $\mathbb{Q}$-combination of the isogeny iterates' graphs – an algebraic correspondence; $P = \mathrm{id}$ on W, $P(\theta^2) = 0$, P annihilates the ambient and mixed sectors.

Proof. The proofs of Rounds 38 (§2–§3) and 39 (§2) are verbatim in d: the letter-pair block argument for (i) never uses the Heegner property; the transversal-eigen calculus for (ii) is the identity $\bigwedge^4(e_j + \sqrt{-d} f_j) = \hat{w}_R + \sqrt{-d}\, \hat{w}_I$ (this round: certified *symbolically* in $\mathbb{Z}[d]$, Theorem 3.3 below, not sampled); the periods (iii) are polynomial in d and are certified as polynomial identities; the CRT projector (iv) needs exactly the coprimality of (i), which holds on the family by Theorem 1.1(iii). The certificate verifies all receipts at the 62-world census by exact 70×70 arithmetic. □

3 The crossing at the family and the K-irreducibility dichotomy

Theorem 3.1 (S31, the crossing transport at the family)**.** *All three discovery layers and the full Round 39 crossing assembly hold verbatim at every family world: the eigenbranch lattice law (the parity iff $m+2w$ even; the born instances $(5,1),(4,1),(7,3)$ refused, $(6,2)$ admitted), the exchange-deck residue law ($H^2(C_2;M)=0$ by norm-surjectivity, the invariant generation exact, the untyped control firing), the cited K-freeness inputs with the exterior/symmetric discrimination executed, and the $(6,2)/(3,3)$ assembly:*

$$q(E_\pm)=(\,24d^2,\ -8d^2\,),\quad c_4(E_\pm)=\left(\tfrac{72d^2+24}{2},\ \tfrac{8d^2+24}{2}\right),\quad \textit{cross}=-(24d^2+24),$$

the carrier $c_4=16d^2=2\,\mathrm{INT}\,\hat w_R^2$, the four faces $(8d^2,16d^2,32d^2)$, the W6 tables in the general-world form $t_1=m^2q-mq_M$ etc., the E1–E4 entry test green, and

$$\boxed{m^*(A_K,2,\hat w_R)=2\ \textit{EXACT at every family world.}}$$

Proof. Round 39's proofs are d-uniform: the branch integrality is the $t=0/t=1$ coordinate-forcing (the coefficient 1 is d-independent); the deck law is world-independent; the charges are the $\hat w_R$-pairings whose values are the periods of Theorem 2.1(iii). The certificate re-executes every receipt at the 62-world census. □

Theorem 3.2 (S32, the K-irreducibility dichotomy – the sharpening)**.** *For every d and every $\beta=c+e\sqrt{-d}\in K$ with $e\neq 0$, the block of $[\beta]^*|_W$ in the $(\hat w_R,\hat w_I)$ basis is*

$$[\beta]^*|_W=\begin{pmatrix}A & B\\ -dB & A\end{pmatrix},\qquad \beta^4=A+B\sqrt{-d},$$

*the regular K-representation of β^4 on $K\cong\langle\hat w_R,\hat w_I\rangle_{\mathbb{Q}}$. Consequently the operator algebra $\{[\beta]^*|_W:\beta\in K\}$ is (a) spanned by the scalars $[q]^*=q^4\,\mathrm{Id}$ together with any single non-scalar block, and (b) **irreducible**: the only $\mathbb{Q}$-subspaces of W stable under $[K]$ are 0 and W. In particular the dichotomy holds at* every *family world, including the order-4 and order-6 phase worlds where* minpoly(α^4) *is linear or Round 38's $[\alpha]$-irreducibility criterion degenerates.*

Proof. The eigenletter identity gives $[\beta]\Omega'=\beta^4\Omega'$ and $[\beta]\Omega''=\bar\alpha^4\Omega''$ with $\Omega'=\hat w_R+\sqrt{-d}\hat w_I$, $\Omega''=\hat w_R-\sqrt{-d}\hat w_I$; solving the 2×2 system gives the displayed block (the certificate verifies the block exactly at every census world). The block is the matrix of multiplication by $\beta^4=A+B\sqrt{-d}$ on K in the basis $(1,\sqrt{-d})$; when $B\neq 0$ its characteristic polynomial $t^2-2At+(A^2+dB^2)$ has discriminant $-4dB^2<0$, hence is irreducible over $\mathbb{Q}$, so a single non-scalar block already admits no stable $\mathbb{Q}$-line. A subspace stable under the full $[K]$-algebra is in particular stable under a non-scalar block (one exists at every d: $\beta_0=1+\sqrt{-d}$ for $d\neq 1$, $2+i$ at $d=1$, since $(1+\sqrt{-d})^4$ has $\sqrt{-d}$-part $4(1-d)\neq 0$ and $(2+i)^4=-7+24i$), whence the subspace is 0 or W. This replaces Round 38/39's $[\alpha]$-dichotomy (irreducible minpoly(α^4)), which fails at the order-4 worlds, with a law that is uniform in d. □

Theorem 3.3 (S33, the leaf and the spread at the family)**.** *At every family world: $P(\hat w_R)=\hat w_R\neq 0$ (the one-source leaf fires); the $[\beta]^*$-translates of $\hat w_R$ (five distinct β-directions, certified) span inside W; and by the K-irreducibility dichotomy, any algebraic $z\in W\setminus\{0\}$ generates W under the algebraic correspondences $[K]$ – one algebraic source closes the full plane. Given the external leaf $W\subset\mathrm{Hdg}^4$ (the committed scope of Rounds 38/39, unchanged),* $\mathrm{Hdg}^4=\mathbb{Q}\theta^2\oplus W$ *is algebraic at every family world: the RHC at degree 4 on the family (inside the expanded frame, at the model level).*

4 The symbolic-d layer (the quantifier layer)

Theorem 4.1 (S34, the polynomial identities)**.** *In $\mathbb{Z}[d][\sqrt{-d}]$ with exact polynomial arithmetic:*

(i) the eigenletter construction $\bigwedge_j^4(e_j + \sqrt{-d}\, f_j) = \hat{w}_R(d) + \sqrt{-d}\,\hat{w}_I(d)$ is an identity of $\mathbb{Z}[d]$-valued classes (the 16 transversal words, the $(1, -d, d^2)/(1, -d)$ coefficients exact);

(ii) $\mathrm{INT}\,\hat{w}_R(d)^2 = 8d^2$, $\mathrm{INT}\,\hat{w}_I(d)^2 = 8d$, $\mathrm{INT}\,\hat{w}_R(d)\hat{w}_I(d) = 0$, $\mathrm{INT}\,\hat{w}_R(d)\,\theta^2 = 0$, $\mathrm{INT}\,\theta^4 = 24$ *– polynomial identities;*

(iii) the branch charges $q(E_\pm) = (24d^2, -8d^2)$, the Chern closure $\big((72d^2 + 24)/2,\ (8d^2 + 24)/2,\ -(24d^2 + 24)\big)$, the carrier $16d^2$, the Z-face $8d^2$ and the B-face $16d^2$ – polynomial identities.

These hold at every *d – the family theorem's arithmetic content is a polynomial identity, not a census of samples. Evaluated at $d = 1..30$ they reproduce the per-world closed forms (the layer bridge).*

Proof. Direct exact computation in the polynomial ring: the certificate's `VSYM` layer implements $\mathbb{Z}[d]$ as dense integer tuples and (r, s)-pairs as $\mathbb{Z}[d][\sqrt{-d}]$, executes the wedge algebra over that ring, and checks each identity against its closed form. The polynomial identity theorem (two polynomials of degree $\leq D$ agreeing at $D + 1$ points are equal) certifies that these are identities rather than coincidences at the census; the certificate additionally checks the d-substitution bridge against the census layer. □

5 The certificate and the scope

`d_family_transport_round40_checks.py`: 4223 checks, 0 failures, exit 0, deterministic double-run byte-identical (post-repair md5 `eed81484d0df4c58a26d7fff43417666`; pre-repair 4219/`acc2455cbe07c0a881`
seed `0x4D9D1F1CA7B0B5E` distinct from every prior certificate); exact Fraction / integer-polynomial arithmetic, zero floats; ∼80 s; the census 62 family worlds + 5 collar canaries; bounded (matrices $\leq 70 \times 70$, parity sweeps $\leq 13 \times 13 \times 5$ per world with the full $19 \times 19 \times 7$ at the discriminator seed, c_1 samples ≤ 150 per world, symbolic degree ≤ 8). Sections: V0 the phase classification (the 160-instance grid, both directions, the collar cross-detected by the coprime machinery); V1 the world-interface law; V2 the projector and the leaf; V3 the eigenbranch lattice law; V4 the exchange-deck residue law (world-independent, re-certified); V5 the word-algebra constants (cited; the exterior/symmetric discrimination computed by two real product operations); V6 the crossing assembly; V7 the pin (the content-bearing leastness scan), the spread, the K-irreducibility dichotomy (the exact block law), the degenerate canaries (the Eisenstein merge dim 34 twice, the order-2 dim 38 twice, the order-1 dim 70 – five canaries); VSYM the symbolic-d polynomial-identity layer with the eigenletter identity, the layer bridge on the *computed* polynomials, and the strong bridge (symbolic = census at all 62 worlds); V8 the hostile battery (the world-mutation canary at non-Heegner worlds, letter-deletion, projector-sign, the input-sensitivity control, garbage liveness, the four-direction parity canaries, the theta-half mutation, the planted misclassification canaries) + the mechanical vacuity scan + the self-comparison scan.

The audit. AUDIT-40 (one fresh adversarial agent, the PTS adversarial-auditor role): verdict PASS_WITH_REPAIRS_REQUIRED; all five external sabotages fired (exit 1); the agent's

independent recompute (its own code, different conventions) at three worlds: 35/35 agreement, 0 disagreements. Six findings – the major one being a vacuous layer-bridge check (hardcoded tuples on both sides, the 092/107 class recurrence) – all six repaired in one batched round strengthen-to-label; the closing re-check 4223/0/exit 0 deterministic; the post-repair rigging re-test on the repaired bridge caught (exit 1). Loop closed.

The scope line (unchanged from Rounds 38/39). The external leaf $W \subset \mathrm{Hdg}^4$ (the Weil-type Hodge-theoretic input, the repo's forcing frame) is *given*, not proved here; the RHC at degree 4 on the family is certified at the same model-level scope as the discriminator's discovery-strike receipt. The letter ring is $\mathbb{Z}[\sqrt{-d}]$ (the maximal order at $d \equiv 1 \bmod 4$ is larger; out of scope, as in all prior rounds). The Clay statement is NOT claimed; the closing string is NOT emitted; the polarity split stands.

The quantifier line (§37)

What this round did to the universal quantifier: the Weil face of the transport family is closed – the nine-world census is revealed as the seed of the infinite family $\mathcal{F}$ (every squarefree d, every α outside the exact collar $\{b = 0\} \cup \{a = 0\} \cup \{(3, a = \pm b)\}$), the package certified at a 62-world census spanning 31 distinct d's with the arithmetic laws certified as polynomial identities in d, the order-4/6 phase collars *added* by the K-irreducibility sharpening, and the leaf firing at every family world. The remaining falsifiable line: the CM-side transport family (the Round-37 census boundary: degree-8 closures of order ≥ 24, degrees ≥ 10), R3 (the K-condition at general input), R4 (the transcendental fold-birth). Q2 NOT claimed; the closing string NOT emitted; the polarity split stands.